Yuantao Wei

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Research Interests

Causal discovery and inference; probabilistic machine learning; LLM-based decision-making.

Education

University of Washington Seattle, WA
Ph.D. in Industrial Engineering Sep 2026 – Present
Advisor: Prof. Shuai Huang
Honors: Top Scholar Award, UW Graduate School (2026) | NeurIPS 2026 Financial Assistance Award
M.S. in Industrial Engineering, GPA: 3.68 Sep 2024 – Jun 2026

Shanghai University Shanghai, China
B.A. in Industrial Engineering, GPA: 3.68 Sep 2020 – Jun 2024

Technical Skills

Programming: Python, SQL, R
Deep Learning Frameworks: PyTorch, TensorFlow, scikit-learn
Causal Modeling Libraries: causal-learn, pgmpy, DoWhy, gCastle
Research tools: Git, GitHub, LaTeX, Cloud computing

Research Experience

University of Washington Seattle, WA
Research with Prof. Shuai Huang

Causal Discovery from Human Knowledge MoC; NeurIPS 2026 [1]

- Developed Mixture-of-Chains (MoC), a causal discovery framework that represents expert judgments through a mixture of latent causal chains. Combined latent chain discovery, aggregation, and DAG decoding in a multi-stage framework to recover causal structure directly from expert knowledge.
- Reduced structural Hamming distance (SHD) by **29% on average** across 5 networks and 4 ranking-based baselines, with a **46% reduction relative to the strongest baseline**. Further evaluated MoC-derived priors for observational causal discovery, improving graph recovery.

Bayesian Modeling of Expert Judgments DAG-IRT; publication [2]

- Developed DAG-IRT, a Bayesian item response framework for aggregating expert judgments to recover causal graphs. Jointly inferred graph structure, expert ability, and judgment difficulty, accounting for differences in expertise while quantifying posterior uncertainty over the recovered graph.
- Outperformed all evaluated crowdsourcing baselines in causal graph recovery across 4 synthetic benchmark networks and 2 human-judgment tasks, achieving **27% lower SHD on average** than majority voting and other crowdsourcing baseline such as Dawid–Skene.

Causal Discovery under Distribution Shifts KRD; publication [3]

- Investigated causal direction from two regimes using kernel residual displacement, with identifiability and finite-sample guarantees under the study's assumptions.
- KRD averaged **10.8 percentage points higher accuracy** than 9 baselines across 8 real-data tasks.

LLM Agents and Digital Twins for Decision Support TwinPlan; publication [4]

- Contributed to an LLM-agent and digital-twin framework that translates personalized health plans into executable interventions and compares them through counterfactual simulation.
- On **100 semi-synthetic cases**, achieved **Set F1 0.900**: a **41% average gain across 3 baseline families** and **22% above the strongest baseline**, with 100% decision coverage.

Digital Twins for Food Supply Chains Ongoing research

- Developing a digital-twin framework for food supply chains, focusing on last-mile cold-chain delivery and the coupling of food-quality dynamics with logistics decisions. Studying adaptive decisions through sequential twin-state updates, considering trade-offs among shelf life, transportation cost and food waste.

Selected Publications

[1] Mixture-of-Chains: Learning Causal Graphs from Human Knowledge
Yuantao Wei*, Huiling Liao*, Ryan F. Lin*, Xiaoning Qian, and Shuai Huang.
Accepted at NeurIPS 2026

[2] Bayesian DAG-IRT: Aggregating Expert Judgments for Causal Graph Recovery
Yuantao Wei*, Huiling Liao*, Ryan Feng Lin, Congjing Zhang, Xiaoning Qian, and Shuai Huang.
Under review at AISTATS 2027

[3] The System Is Out of Control, but Causality Isn't: Leveraging Operational Shifts for Bivariate Causal Discovery
Keyu Tian, Yuantao Wei, Shuai Huang, and Li Zeng.
Under review at ICLR 2027

[4] TwinPlan: Evidence-Supported Personalized Health Planning via LLM Agents and Digital-Twin Simulation
Chunli Peng, Ryan Feng Lin, Siyuan Xu, Wei-Chien Kao, Congjing Zhang, Yuantao Wei, Jundi Liu, Huiling Liao, Shuai Huang, and Xiaoning Qian.
Under review at KDD 2027

[5] Causal Learning Should Embrace the Wisdom of the Crowd
Ryan F. Lin*, Yuantao Wei*, Huiling Liao, Xiaoning Qian, and Shuai Huang.
Position paper | arXiv:2603.02678

*Equal contribution.

Industry Experience

Foresight Fund Management Company Shanghai, China
Quantitative Research Intern Mar 2024 – Aug 2024

- Replicated and refined A-share strategies, including equal-weight rebalancing and QMJ variants; built a systematic backtesting framework and analyzed discrepancies between backtest and live-market performance.

Deloitte Shanghai, China
Data Analysis Intern Sep 2023 – Dec 2023

- Built Python pipelines for data collection, cleaning, and integration to support model-training workflows for an intelligent data platform.
- Assisted with model testing, prompt tuning, and deployment validation for LLM and multimodal-generation applications.

MicroPort Medical
Algorithm Intern Jul 2023 – Sep 2023

- Developed a CNN-based sound-scene classification workflow for cochlear-implant processors using **40,000+ audio samples**; improved accuracy from **60% to 85% (+25 percentage points)** through architecture optimization and test refactoring.
- Prepared technical documentation and supported patent-material preparation.

Teaching

Teaching Assistant, University of Washington
IND E 527: Data Analytics for Systems Engineering Fall 2026
IND E 321: Statistical Quality Control Spring 2025